Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

Yong-Zher Rui

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Abstract

We extend a two-stage stochastic virtual bus stop (VBS) selection model to handle uncertain passenger demand within each scenario type (morning, afternoon, evening, night). Demand for each origin–destination pair (o, d) and scenario type s is modeled as lying in a capped-total-demand uncertainty set parameterized by per-OD upper bounds together with a per-scenario upper bound on aggregate demand; lower bounds are fixed at zero by design. We derive the dual robust counterpart of the inner maximization, which remains linear and can be solved with the same MILP machinery as the nominal model. We compare the nominal base model with its robust total-demand-cap counterpart on out-of-sample demand realizations and report average cost, high-percentile (P90/P95) cost, and the price of robustness.

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1 Introduction

Microtransit systems offer flexible, on-demand shared rides between a set of designated virtual bus stops (VBS). A key design decision is which candidate locations to build as VBS. Prior work formulates this as a two-stage stochastic mixed-integer program: in the first stage, build l stations and choose which $k \leq l$ are active in each scenario $s \in S$; in the second stage, assign passengers to origin–destination VBS pairs.

A limitation of the stochastic model is that it treats the demand q_{ods} within each scenario as a fixed, known quantity (typically the historical mean). In practice, demand fluctuates even within a scenario type: morning-peak demand on one day may differ substantially from the next. This motivates a *robust* extension where q_{ods} is uncertain within a bounded set.

The uncertainty model adopted here is designed to encode a direct operational statement: each OD pair has a plausible demand range, and the *total* demand within a scenario type should not exceed a calibrated upper bound. This produces a tractable robust counterpart for the base station-selection model.

Accordingly, the report focuses on two formulations only:

- the nominal base model, which optimizes against scenario-average demand;
- the robust counterpart, which protects against worst-case demand realizations within a capped uncertainty set.

Under a nominal-demand optimization, the chosen station set may fit the average demand pattern well but perform poorly when demand shifts toward OD pairs that are less conveniently located relative to the selected stops. The robust model reduces this sensitivity by favoring station configurations whose assignment performance remains more stable across plausible demand realizations.

2 Base Model

2.1 Decision variables

$y_j \in \{0,1\}$	build station j
$z_{js} \in \{0,1\}$	station j is active in scenario s
$x_{odjks} \in \{0,1\}$	assign OD pair (o,d) to origin VBS j, destination VBS k in scenario s

Following the March 18 JacqWang formulation, walking distance induces admissibility sets. Let $\Delta(o) \subseteq J$ be the candidate pickup stations within the walking limit of origin o, and let $\Delta(d) \subseteq J$ be the candidate dropoff stations within the walking limit of destination d. Define the admissible station-pair set

$$A_{od} := \{(j,k) : j \in \Delta(o),\ k \in \Delta(d),\ j \neq k\}.$$

Only pairs in A_{od} may be used to serve OD pair (o,d).

2.2 Nominal objective

The nominal two-stage objective minimizes expected passenger travel cost:

$$\min_{x,y,z} \sum_{s\in S} \sum_{(o,d)\in\Omega} \sum_{(j,k)\in A_{od}} q_{ods} \underbrace{\left(d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda\, c_{jk}\right)}_{a_{odjks}} x_{odjks}, \tag{1}$$

where d_{oj}^{orig} is the walking distance from origin o to VBS j, d_{dk}^{dest} is the walking distance from VBS k to destination d, c_{jk} is the in-vehicle travel time between VBS j and k, and λ is a weight that converts routing time to the same unit as walking distance.

2.3 Constraints

$$\sum_{(j,k)\in A_{od}} x_{odjks} = 1 \qquad \forall (o,d), s \tag{2}$$

$$x_{odjks} \le z_{js}, \quad x_{odjks} \le z_{ks} \qquad \forall (o,d), (j,k) \in A_{od}, s \tag{3}$$

$$z_{js} \le y_j \qquad \forall j, s \tag{4}$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \qquad \forall s \tag{5}$$

3 Uncertainty Set for Demand

3.1 Parameterization

For each OD pair (o,d) and scenario type s, let $\overline{q}_{ods}$ denote an upper bound on demand, and let $\overline{Q}_s$ denote an upper bound on *total* demand in scenario type s. Lower bounds are fixed at zero (see §6.2 for justification). The uncertainty set is

$$\mathcal{U}_s = \Big\{ q_s : 0 \le q_{ods} \le \overline{q}_{ods}\ \forall (o,d), \sum_{(o,d)\in\Omega} q_{ods} \le \overline{Q}_s \Big\}. \tag{6}$$

This uncertainty set says two things simultaneously:

- each OD pair must remain within its own historically calibrated range;
- the aggregate demand in the scenario cannot exceed $\overline{Q}_s$.

3.2 Interpretation of the total-demand cap

- If $\overline{Q}_s = 0$, then all OD demands are forced to zero (vacuous).
- If $\overline{Q}_s \ge \sum_{(o,d)} \overline{q}_{ods}$, then the total demand cap is inactive and the model reduces to independent box bounds on each OD pair.
- Intermediate values of $\overline{Q}_s$ allow some OD pairs to be high simultaneously, but only so long as total demand stays below the calibrated cap.

A larger $\overline{Q}_s$ for a high-demand period such as evening encodes the prior that those periods experience larger systemwide demand surges.

3.3 Remark on integer demand

Observed demand is integer-valued, but the robust inner maximization may still be solved as a linear program. The key point is that the uncertainty set (6) is a bounded knapsack-type polytope. If the bounds $\overline{q}_{ods}$ and $\overline{Q}_s$ are integral, the inner LP has an optimal extreme point that is integral as well. Hence dropping integrality in the inner maximization does not change the worst-case value, even though the uncertainty set is written in continuous form.

4 Robust Counterpart

4.1 Robust objective

The robust model replaces the nominal objective with

$$\min_{y,z} \sum_{s\in S} \max_{q_s\in\mathcal{U}_s} \Phi_s(z_s, q_s), \tag{7}$$

subject to the first-stage constraints (4)–(5), where $\Phi_s(z_s, q_s)$ is the recourse value defined below.

4.2 Recourse value function

The key modeling point is that x is a *recourse* decision. For fixed scenario-active stations z_s and realized demand vector q_s, define the recourse value function

$$\Phi_s(z_s, q_s) := \min_x \sum_{(o,d)\in\Omega} \sum_{(j,k)\in A_{od}} q_{ods}\, a_{odjks}\, x_{odjks} \tag{8}$$

subject to

$$\sum_{(j,k)\in A_{od}} x_{odjks} = 1 \qquad \forall (o,d), \tag{9}$$

$$x_{odjks} \le z_{js}, \quad x_{odjks} \le z_{ks} \qquad \forall (o,d), (j,k) \in A_{od}, \tag{10}$$

with $x_{odjks} \ge 0$. Integrality is not needed here: for fixed (o, d, s) the feasible region is a simplex intersected with bound constraints, so an optimal extreme point selects a single active station pair.

For fixed z_s, this problem decomposes by OD pair. Define the induced minimum assignment cost

$$m_{ods}(z_s) := \min_{x_{od\cdot\cdot s}} \left\{ \sum_{(j,k)\in A_{od}} a_{odjks}\, x_{odjks} : \sum_{(j,k)\in A_{od}} x_{odjks} = 1,\ x_{odjks} \le z_{js},\ x_{odjks} \le z_{ks}\ \forall (j,k) \in A_{od} \right\}. \tag{11}$$

Because all demand coefficients are nonnegative and the recourse constraints are separable across OD pairs, the recourse value can be written as

$$\Phi_s(z_s, q_s) = \sum_{(o,d)\in\Omega} q_{ods}\, m_{ods}(z_s). \tag{12}$$

For fixed z_s, the inner worst-case demand problem in (7) becomes

$$\max_{q_s\in\mathcal{U}_s} \sum_{(o,d)} q_{ods}\, m_{ods}(z_s) \quad \text{s.t. } 0 \le q_{ods} \le \overline{q}_{ods},\ \sum_{(o,d)} q_{ods} \le \overline{Q}_s. \tag{13}$$

This is a continuous knapsack problem, most conveniently handled by duality.

4.3 Derivation of the robust counterpart

Since $\underline{q}_{ods} = 0$ by design, $B_s := \overline{Q}_s$ exactly, and the inner maximization (13) simplifies to a continuous knapsack:

$$\max_{r \geq 0} \sum_{(o,d)} m_{ods}(z_s)\, r_{ods} \quad \text{s.t.} \quad r_{ods} \leq \overline{q}_{ods}\ \forall (o,d), \quad \sum_{(o,d)} r_{ods} \leq \overline{Q}_s. \tag{14}$$

Why $m_{ods}(z_s)$ can be treated as a constant. The duality is applied only to the *inner* LP, for a **fixed** z_s. With z_s held fixed, the set of active stations is known, and $m_{ods}(z_s)$ (11) is a deterministic scalar — the minimum feasible assignment cost for OD pair (o,d) given those stations. The inner LP then has r as its only variable and $m_{ods}(z_s)$ as an ordinary constant objective coefficient. LP duality applies to this fixed-z_s inner LP, producing dual constraints that contain $m_{ods}(z_s)$ as a parameter. It is only when we re-introduce the outer minimization over (y,z) that $m_{ods}(z_s)$ becomes a function of a decision variable; the x-variable substitution in Step 3 handles exactly this.

Step 1: Identify dual variables from the primal constraints. The inner LP (14) has two families of $\leq$ constraints. LP duality assigns one non-negative dual variable to each:

Primal constraint	RHS	Dual variable
$\sum_{(o,d)} r_{ods} \leq \overline{Q}_s$	$\overline{Q}_s$	$\alpha_s \geq 0$
$r_{ods} \leq \overline{q}_{ods}$, each (o,d)	$\overline{q}_{ods}$	$\beta_{ods} \geq 0$

Step 2: Build the dual objective and constraints. For a maximization LP ($\max c^T r$, $Ar \leq b$, $r \geq 0$), the dual is $\min b^T y$ subject to $A^T y \geq c$, $y \geq 0$.

Dual objective. Each primal constraint contributes its RHS times its dual variable:

$$\min \quad \overline{Q}_s\, \alpha_s \ + \sum_{(o,d)} \overline{q}_{ods}\, \beta_{ods}.$$

Dual constraint for each primal variable r_{ods}. Variable r_{ods} has objective coefficient m_{ods} in the primal. It appears with coefficient 1 in the budget constraint (dual: α_s) and coefficient 1 in its own box constraint (dual: β_{ods}), and zero in every other box constraint. The dual constraint for r_{ods} is therefore:

$$1 \cdot \alpha_s \ + \ 1 \cdot \beta_{ods} \ \geq \ m_{ods} \qquad \Longrightarrow \qquad \alpha_s + \beta_{ods} \geq m_{ods}(z_s).$$

Step 3: Dual LP and strong duality. Collecting the above, the dual of (14) is:

$$\min_{\alpha_s \geq 0,\ \beta_{ods} \geq 0} \quad \overline{Q}_s\, \alpha_s + \sum_{(o,d)} \overline{q}_{ods}\, \beta_{ods} \qquad \text{s.t.} \quad \alpha_s + \beta_{ods} \geq m_{ods}(z_s) \quad \forall (o,d). \tag{15}$$

Strong duality holds: the primal (14) is feasible ($r = 0$) and bounded (finite objective on a bounded polytope), so the optimal values of the primal and dual coincide.

Step 4: Embed the recourse minimization. The dual constraint (15) still contains the recourse value $m_{ods}(z_s)$ from (11), which is itself a minimization over assignment variables x. We substitute directly: introduce continuous assignment witnesses $x_{odjks} \geq 0$ satisfying (9)–(10), and replace $m_{ods}(z_s)$ with $\sum_{(j,k)\in A_{od}} a_{odjks}\, x_{odjks}$. The constraint becomes:

$$\alpha_s + \beta_{ods} \;\geq\; \sum_{(j,k)\in A_{od}} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s. \tag{16}$$

Note that no auxiliary t_{ods} variable is needed: (16) directly links the dual variables to the assignment cost of x.

For any feasible x, the right-hand side $\sum_{(j,k)\in A_{od}} a_{odjks} x_{odjks}$ is an upper bound on $m_{ods}(z_s)$, so (16) is at least as tight as the original dual constraint. At optimality the minimization of $\overline{Q}_s \alpha_s + \sum \overline{q}_{ods} \beta_{ods}$ drives each $x_{od\cdot\cdot s}$ to minimize its assignment cost — a smaller right-hand side relaxes (16), allowing smaller (α_s, β_{ods}), which reduces the objective. Hence the optimal x achieves $\sum_{(j,k)\in A_{od}} a_{odjks} x_{odjks} = m_{ods}(z_s)$, and the two formulations are equivalent.

Robust counterpart MILP. Combining with the outer minimization over (y, z), the full formulation is:

$$\min_{x,y,z,\alpha,\beta} \sum_s \overline{Q}_s\, \alpha_s + \sum_s \sum_{(o,d)} \overline{q}_{ods}\, \beta_{ods} \tag{17}$$

subject to

$$\alpha_s + \beta_{ods} \geq \sum_{(j,k)\in A_{od}} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{18}$$

$$\sum_{(j,k)\in A_{od}} x_{odjks} = 1 \qquad \forall (o,d),\ s, \tag{19}$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \qquad \forall (o,d),\ (j,k) \in A_{od},\ s, \tag{20}$$

plus (4)–(5), with $x_{odjks}, \alpha_s, \beta_{ods} \geq 0$.

5 JuMP Implementation Sketch

The data required for the robust model (in addition to the base model data) are:

```
q_high[od, s]      # per-OD upper demand bound  (q_bar_ods)
Q_cap[s]           # total-demand cap            (Q_bar_s)
a[od, j, k, s]     # assignment cost
```

New variables (x is *continuous*—a recourse witness, not a demand count):

```
@variable(model, x[od in OD, j in J, k in J, s in S] >= 0)
@variable(model, alpha[s in S] >= 0)
@variable(model, beta[od in OD, s in S] >= 0)
```

Robust objective:

```
@objective(model, Min,
    sum(Q_cap[s] * alpha[s] for s in S)
    + sum(q_high[od,s] * beta[od,s] for od in OD, s in S)
)
```

Assignment witness constraints ($\sum_{j,k} x = 1$, $x \le z$):

```
@constraint(model, [od in OD, s in S],
    sum(x[od,j,k,s] for j in J, k in J) == 1)
@constraint(model, [od in OD, j in J, k in J, s in S],
    x[od,j,k,s] <= z[j,s])
@constraint(model, [od in OD, j in J, k in J, s in S],
    x[od,j,k,s] <= z[k,s])
```

Robust dual constraint (no auxiliary t needed—assignment cost appears directly):

```
@constraint(model, [od in OD, s in S],
    alpha[s] + beta[od,s] >=
        sum(a[od,j,k,s] * x[od,j,k,s] for j in J, k in J)
)
```

The first-stage constraints (4)–(5) are unchanged.

6 Experimental Setup

We evaluate the model on the Zhuzhou microtransit dataset, which contains 89,669 trips recorded over 111–114 days per time-of-day period. Trips are partitioned into four scenario types by start hour: morning (06–10), afternoon (10–15), evening (15–20), and night (20–24). The calibration script `scripts/generate_demand_bounds.jl` reads the historical order data, groups it by period, computes per-OD quantiles and a total-demand cap, and writes the bounds to `Data/demand_bounds.json`.

6.1 Calibrating the uncertainty set

Group historical observations by scenario type $s \in \{\text{morning}, \text{afternoon}, \text{evening}, \text{night}\}$. For each OD pair (o, d), set the lower bound to zero and compute the upper bound as an empirical quantile:

$$\underline{q}_{ods} = 0, \qquad \overline{q}_{ods} = Q_{q_{\text{high}}}\big(q_{od}^{\text{hist}} \mid s\big). \tag{21}$$

The rationale for fixing $\underline{q}_{ods} = 0$ is discussed in §6.2.

Next compute the realized total demand on each historical day:

$$Q_{s,t}^{\text{hist}} := \sum_{(o,d)\in\Omega} q_{ods,t}^{\text{hist}}, \tag{22}$$

where t indexes historical observations within scenario type s. Then set

$$\overline{Q}_s = Q_{Q_{\text{cap}}}\big(Q_{s,t}^{\text{hist}}\big), \tag{23}$$

where Q_{cap} is the chosen calibration percentile (e.g., 0.90).

One practical check is to verify that

$$\sum_{(o,d)} \underline{q}_{ods} \le \overline{Q}_s \le \sum_{(o,d)} \overline{q}_{ods};$$

the left inequality holds trivially since $\underline{q}_{ods} = 0$, and the right inequality requires that the chosen $\overline{Q}_s$ not exceed the sum of all per-OD upper bounds. When it does, the cap is non-binding and the robust model degenerates to a pure box model.

6.2 Empirical analysis of the uncertainty set

Lower bounds: $\underline{q}_{ods} = 0$. We fix the lower bound at zero for all OD pairs, for two reasons. First, demand is highly sparse: most OD pairs appear on fewer than 10% of days, so a quantile-based lower bound would be zero for the vast majority of pairs regardless. Second, and more importantly, $\underline{q} = 0$ is the *conservative* choice for the operator. In the robust formulation, $\underline{q}_{ods}$ acts as a floor that *constrains the adversary from below* — it forces them to place at least $\underline{q}_{ods}$ demand on every OD pair, even cheap ones, consuming budget that could otherwise be concentrated on the most costly corridors. Setting $\underline{q} = 0$ removes this floor and allows the adversary maximum flexibility, giving the operator stronger protection. The practical consequence is that $B_s = \overline{Q}_s$ exactly, and the $\sum \underline{q} \cdot t$ term in the robust objective vanishes entirely.

Upper bounds: $\overline{q}_{ods} = Q_{0.90}$ of zero-padded historical demand. The total-demand cap $\overline{Q}_s$ is the 90th percentile of the daily aggregate demand in period s.

Is the total-demand cap binding? The central question is whether $\sum_{(o,d)} \overline{q}_{ods} > \overline{Q}_s$. If not, the cap is slack even when every OD pair hits its maximum simultaneously, and the robust model degenerates to a pure box model. Formally:

- **Non-binding (pathological):** $\sum \overline{q}_{ods} \leq \overline{Q}_s$ — the budget is never exhausted; robustness is entirely determined by per-OD bounds.
- **Binding (meaningful):** $\sum \overline{q}_{ods} > \overline{Q}_s$ — the adversary cannot push all OD pairs to their upper bounds at once.

Table 1 reports the key quantities; all four periods are in the binding regime.

Period	$\sum \underline{q}$	$\sum \overline{q}$	$\overline{Q}$	gap $\sum \overline{q} - \overline{Q}$	ratio $\frac{\sum \overline{q}}{\overline{Q}}$	active OD pairs
Morning	0.0	173.0	110.0	+63.0	1.57	102 / 901
Afternoon	2.0	503.7	376.9	+126.8	1.34	221 / 1589
Evening	15.2	936.2	695.6	+240.6	1.35	319 / 1802
Night	2.0	255.7	208.5	+47.2	1.23	121 / 851

Table 1: Uncertainty set statistics per period. The gap $\sum \overline{q} - \overline{Q}$ is the total demand "removed" from the box by the cap; the ratio > 1 confirms the cap is binding. "Active OD pairs" are those with $\hat{q}_{ods} > 0$ (appear on $\geq$10% of days) out of all observed OD pairs.

Figure 1 shows the three landmarks $\sum \underline{q}$, $\overline{Q}$, and $\sum \overline{q}$ on a common axis per period. The colored segment is the feasible range of total demand under the cap; the faded extension to $\sum \overline{q}$ is the additional demand the *box* would allow but the cap removes.

What does the adversary do? The cap forces the adversary to trade off demand across OD pairs. To see this, rank active OD pairs by $\overline{q}_{ods}$ descending and compute the cumulative sum. Once this cumulative sum reaches $\overline{Q}_s$, the budget is exhausted and the adversary must leave remaining OD pairs at their lower bounds. Figure 2 shows this greedy-filling picture for all four periods.

Historical total demand vs. the cap. Figure 3 shows the realized total demand on each historical day alongside the cap $\overline{Q}_s$ and the box ceiling $\sum \overline{q}$. Because $\overline{Q}_s = Q_{0.90}$(historical total), approximately 10% of days fall above the cap line (measured: 9.9–10.7%). Historical demand never

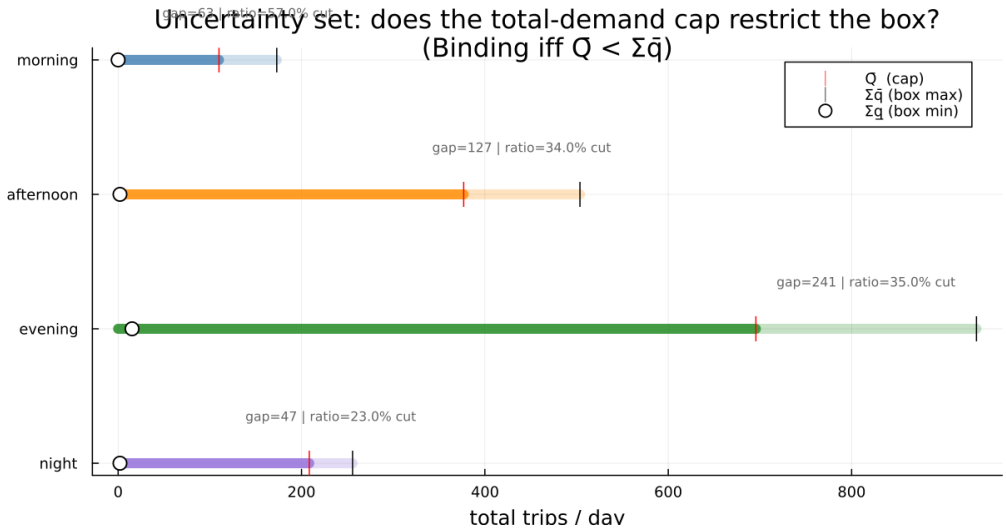

Figure 1: Budget number line per period. The solid colored segment spans $[0, \overline{Q}]$ (feasible under the cap); the faded extension reaches $\sum \overline{q}$ (the box maximum). The "gap" annotation shows how much total demand the cap cuts relative to the box. All periods have gap > 0, confirming the cap is non-redundant.

reaches $\sum \overline{q}$ — this confirms that the box maximum is a theoretical extreme that does not occur in practice, and the cap provides a tighter, data-grounded bound.

What the model hedges against. Figure 4 compares, for the top-30 OD pairs by $\overline{q}$, the adversary's worst-case greedy allocation against the median historical demand per pair. On a typical day most pairs are well below their upper bounds; the adversary concentrates budget into the highest-$\overline{q}$ pairs. This illustrates that the robust model is not hedging against every OD pair spiking at once (impossible under the cap) but against a targeted surge in the most consequential corridors.

The cap gives a reasonable, data-grounded uncertainty set. The key role of the cap is to ensure the uncertainty set is neither trivially small nor unrealistically large. Without the cap, the adversary could push every OD pair to its individual upper bound simultaneously, yielding a total demand of $\sum \overline{q}$ — a level that is never observed historically (Fig. 3, black dotted line). Setting $\overline{Q}_s = Q_{0.90}(\text{total demand})$ constrains the adversary to demand levels that are consistent with what actually occurs on high-demand days, making the uncertainty set *empirically plausible* rather than a purely theoretical worst case.

The calibration percentile governs how conservatively this cap is placed:

Period	$Q_{0.90}$	$Q_{0.95}$	$Q_{0.99}$	$\sum \overline{q}$ (box max)
Morning	110	149	291	173
Afternoon	377	407	543	504
Evening	696	870	1055	936
Night	209	235	403	256

All three quantile choices sit strictly below $\sum \overline{q}$ (except morning where $Q_{0.99} > \sum \overline{q}$, meaning a 99th-percentile cap would be non-binding). The default $Q_{0.90}$ is a reasonable starting point: it

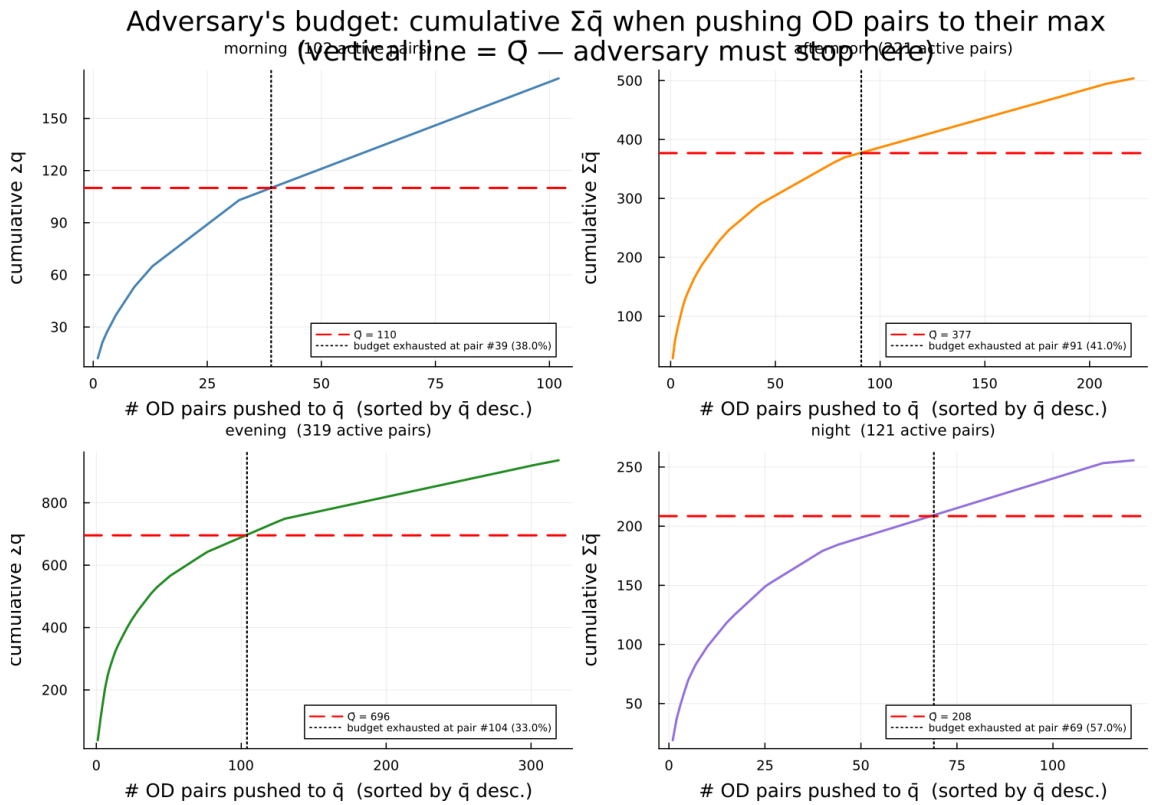

Figure 2: Adversary budget allocation per period. The x-axis counts OD pairs pushed to their upper bound, sorted by $\overline{q}$ descending. The y-axis is the cumulative demand consumed. The red dashed horizontal line is $\overline{Q}_s$; the black dotted vertical line is where the budget is exhausted. Only a subset of all active OD pairs can simultaneously be at their maximum.

corresponds to a high but historically realistic demand level, and the gap $\sum \overline{q} - \overline{Q}_s$ quantifies how much the cap tightens the box. Section 6.4 discusses the expected effects of varying this choice; the main experiment fixes $Q_{\text{cap}} = 0.90$ and $q_{\text{high}} = 0.90$.

Structural observations.

1. **Lower bounds are zero by design**: $B_s = \overline{Q}_s$ exactly, and the $\sum \underline{q} \cdot t$ term in the robust objective vanishes, leaving $\overline{Q}_s \, \alpha_s + \sum \overline{q}_{ods} \, \beta_{ods}$.

2. **Only 11–18% of OD pairs are active**: those present on ≥10% of days receive $\overline{q}_{ods} > 0$; the rest are excluded and assigned zero demand in the worst case. Rare high-demand spikes on sporadic pairs are not hedged.

3. **The cap is genuinely binding**: ratios 1.23–1.57× confirm the set is not pathological.

6.3 Two model variants

We compare:

1. **Nominal model.** Uses $q_{ods} = \mathbb{E}[q_{ods}]$ (scenario-average historical demand).

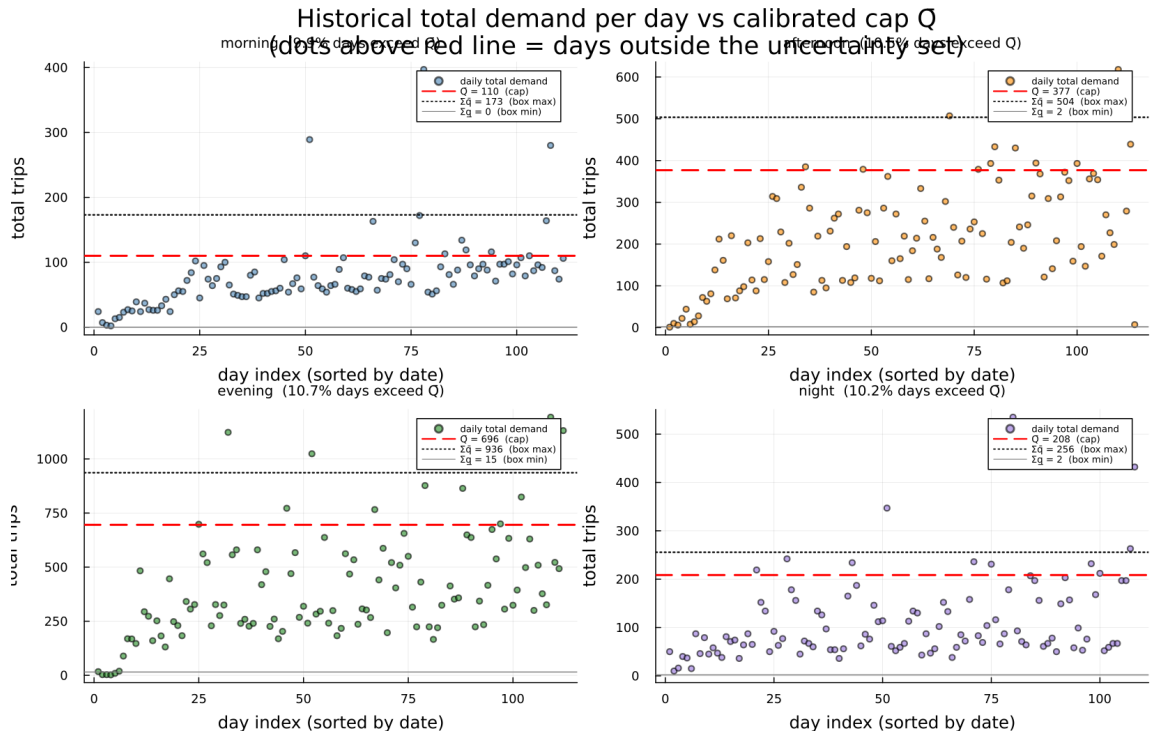

Figure 3: Historical total demand per day (dots) versus the calibrated cap $\overline{Q}$ (red dashed) and the box ceiling $\sum \overline{q}$ (black dotted) per period. The cap sits within the historical range; the box ceiling lies above every observed day.

2. **Robust total-demand-cap model.** Uses uncertainty set $\mathcal{U}_s$ with $\overline{Q}_s$ calibrated from data via (23).

Each model produces a station selection $(y^\star, z^\star)$. Given those station decisions and a realized demand vector, assignment is handled through the recourse problem (8)–(10).

6.4 Sensitivity to calibration quantiles

The uncertainty set has two independent quantile parameters:

- q_{high}: the quantile used for per-OD upper bounds $\overline{q}_{ods} = Q_{q_{\text{high}}}(\text{historical OD demand})$. Controls how wide the individual OD ranges are.
- Q_{cap}: the quantile used for the total-demand cap $\overline{Q}_s = Q_{Q_{\text{cap}}}(\text{historical total demand})$. Controls how tightly the total demand budget is set.

Both are currently set to 0.90 by default (as used in the empirical analysis of §6.2). We sweep both independently to understand how each dimension of conservatism affects the solution:

$$q_{\text{high}} \in \{0.90,\ 0.95,\ 0.99\}, \qquad Q_{\text{cap}} \in \{0.90,\ 0.95,\ 0.99\}.$$

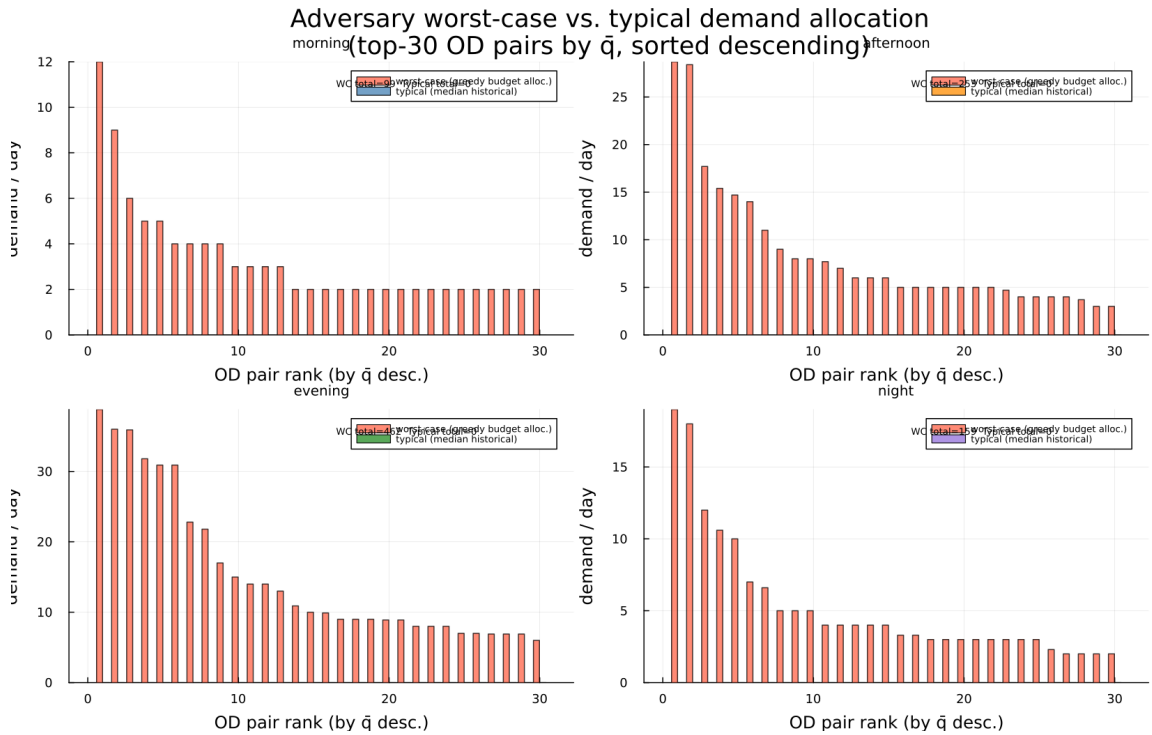

Figure 4: Worst-case adversary allocation (red) vs. typical (median) historical demand (colored) for the top-30 OD pairs by $\overline{q}$ per period. The adversary concentrates budget into the dominant pairs; typical demand is spread more uniformly and is lower in aggregate.

Cap / OD bound quantile		$\overline{Q}_s$ (total-demand cap)			
Q_{cap}	q_{high}	Morning	Afternoon	Evening	Night
0.90	0.90	110	377	696	209
0.95	0.90	149	407	870	235
0.99	0.90	291	543	1055	403
$\sum \overline{q}$ at $q_{\text{high}} = 0.90$		173	504	936	256
$\sum \overline{q}$ at $q_{\text{high}} = 0.95$		—	—	—	—

The expected effects:

- **Raising** Q_{cap}: larger $\overline{Q}_s \to$ adversary has more total-demand budget $\to$ worst-case cost rises $\to$ more conservative design, higher in-sample cost, stronger out-of-sample tail guarantee.
- **Raising** q_{high}: wider per-OD intervals $\to \sum \overline{q}$ increases $\to$ the gap $\sum \overline{q} - \overline{Q}_s$ grows, so the total-demand cap does *more* work relative to the box. Effectively, the adversary has a larger individual budget per corridor.
- **Joint effect**: raising both simultaneously yields the most conservative model; the experiment isolates each dimension.

For the implemented May 2025 study on the full Zhuzhou network, we fix $l = 50$ and use the existing four-period scenario profile. We cross the two model variants with three active-station budgets $k \in \{20, 30, 40\}$, six in-vehicle-time weights $\lambda \in \{0, 0.1, 1, 10, 100, 1000\}$, two walking limits

$\{300, 600\}$ m, and the three-by-three quantile grid above. The full solve grid therefore has $2 \times 3 \times 6 \times 2 \times 3 \times 3 = 648$ configurations. For the nominal model, Q_{cap} and q_{high} are recorded in the job config but have no effect on the solve; the resulting redundancy makes it straightforward to isolate each robustness dimension when analysing results.

6.5 In-sample backtest protocol

We evaluate the two models through an *in-sample backtest*. In the implemented experiment, the calendar window is fixed to May 1, 2025 through May 31, 2025. We use that same month both to calibrate the nominal demand inputs and robust uncertainty sets and to replay realized demand day by day. For each parameter configuration, we solve the nominal and robust models once to obtain designs $(y^\star, z^\star)$. Here $y^\star$ is fixed for the whole month, while the active-station plan is scenario-specific: for each scenario type $s \in S$ there is a corresponding station set $z_s^\star$.

The month is then replayed day by day and period by period. For each realized day-period instance, first identify its scenario type $s^{\text{test}} \in S$ and realized demand vector $q^{\text{test}} = (q_{od}^{\text{test}})_{(o,d)\in\Omega}$. The evaluation then uses the corresponding active-station plan $z_{s^{\text{test}}}^\star$ for that period. In other words, the backtest *rotates through* the precomputed scenario-specific station sets according to the time period being evaluated, while keeping the global build decision $y^\star$ fixed.

For each such realized day-period instance, one need not re-solve a large optimization model. Because the reduced recourse problem is separable across OD pairs and has no demand-coupled capacity constraints, the active station plan $z_{s^{\text{test}}}^\star$ induces a deterministic lookup table: for each OD pair (o, d), pick the cheapest admissible active pair $(j_{od,s}^\star, k_{od,s}^\star) \in A_{od}$. Equivalently, precompute the minimum assignment cost table $t_{od,s}^\star = m_{ods}(z_s^\star)$. The realized cost is therefore

$$\text{Cost}(y^\star, z^\star, s^{\text{test}}, q^{\text{test}}) = \Phi_{s^{\text{test}}}(z_{s^{\text{test}}}^\star, q^{\text{test}}). \tag{24}$$

Operationally, evaluation can therefore be implemented as a table lookup rather than as a repeated LP solve.

Equivalently, one may evaluate

$$\sum_{(o,d)} q_{od}^{\text{test}}\, t_{od,s^{\text{test}}}^\star,$$

where $t_{od,s^{\text{test}}}^\star$ denotes the minimum assignment cost induced by the fixed active stations for the realized scenario type, with the corresponding pair $(j_{od,s}^\star, k_{od,s}^\star)$ restricted to lie in the admissible walking set A_{od}. The total backtest score for the month is obtained by summing or averaging (24) over all realized day-period instances in the chosen calendar window.

This direct cost calculation should be complemented by a simulator-based backtest on the same calendar window. In that second layer, the same fixed design $(y^\star, z^\star)$ is replayed in the operational simulator, again rotating through the scenario-specific station sets $z_s^\star$ by time period. The transformed monthly order file is further split into one file per calendar day, and the simulator runs one replay job per day. The direct lookup-based evaluation measures the model-implied assignment cost, while the simulation backtest measures how the design performs under the full operational dynamics of the simulator. Both are reported side by side rather than collapsed into a single score.

6.6 Reported metrics

For each model, report over the test set:

- Mean realized cost

- 90th and 95th percentile realized cost
- Worst-case realized cost
- Number of OD pairs poorly assigned (realized walking distance exceeding a threshold)
- Stability of station selection across demand realizations
- *Price of robustness*: $\dfrac{\text{Robust obj} - \text{Nominal obj}}{\text{Nominal obj}}$

6.7 Expected finding

The robust total-demand-cap model protects against systemwide demand surges without assuming that every OD pair simultaneously reaches its upper bound. It should improve high-percentile out-of-sample performance relative to the nominal model, while typically accepting some increase in mean in-sample cost.

Formally:

$$\begin{aligned} \text{Nominal P90} &\geq \text{Robust-Cap P90}, \\ \text{Nominal mean} &\leq \text{Robust-Cap mean}. \end{aligned}$$

6.8 Experiment pipeline

The experiment is driven by scripts in `scripts/`:

1. `scripts/generate_demand_bounds.jl`: reads historical order data, groups by time-of-day period, calibrates per-OD quantiles and total-demand caps, and writes `Data/demand_bounds.json`.
2. `scripts/01_setup.jl`: reads the experiment base config and sweep config, then generates per-job TOML configs.
3. `scripts/02_submit.sh`: SLURM array job template.
4. `scripts/03_run_job.jl`: loads config, runs the appropriate model, exports the solved assignment and station decisions, prepares the transformed May 2025 orders, runs the daily simulation replays, aggregates direct-cost and simulator metrics, and saves `metrics.json`.
5. `scripts/04_prepare_backtest.jl`: materializes the month-level transformed orders, daily transformed order files, station and vehicle simulator inputs, and the per-day simulation manifest.
6. `scripts/05_run_simulation_day.jl`: executes one simulator replay job from a single manifest entry.
7. `scripts/06_aggregate_backtest.jl`: combines daily simulator outputs into month-level replay metrics.
8. `scripts/aggregate.jl`: loads all `metrics.json` files, produces `analysis/results.csv` and prints summary statistics including the price of robustness.

The sweep in `experiments/2026-05-01_nominal_vs_robust/` crosses the two model variants, three station budgets $k \in \{20, 30, 40\}$, six in-vehicle-time weights $\lambda \in \{0, 0.1, 1, 10, 100, 1000\}$, two walking limits $\{300, 600\}$ m, and the three-by-three $(Q_{\text{cap}}, q_{\text{high}}) \in \{0.90, 0.95, 0.99\}^2$ quantile grid, giving $2 \times 3 \times 6 \times 2 \times 3 \times 3 = 648$ solved configurations in total. Each solved configuration is then replayed on the full May 2025 month, producing one transformed month-level order file and 31 daily simulator jobs.

7 Conclusion

We propose a robust extension of the VBS station selection model based on per-OD upper bounds together with a scenario-level cap on total demand, with lower bounds fixed at zero by design. The key insight is that, once assignment is written correctly as a recourse function, the resulting worst-case demand problem is a continuous knapsack in demand variables, with a clean dual that yields a linear robust counterpart solvable by the same MILP solver as the nominal model. The parameters $\overline{q}_{ods}$ and $\overline{Q}_s$ have direct empirical interpretations and can be calibrated transparently from historical data. Framed this way, the report isolates the core comparison of interest: the nominal base model versus a tractable robust counterpart that hedges against adverse but bounded demand shifts.

References

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